

BurpTUI

Terminal Interception Proxy — UI, Menu & Feature Design

This document defines the complete navigation structure for BurpTUI: every top-level tab, sub-tab, panel, and feature, organized by build phase. It follows a **phased design** — the interface is designed end-to-end across all three phases, but Phase 2 and Phase 3 tabs/features are shown as **LOCKED** until their corresponding build milestone is reached, matching the project's own incremental build plan (do not advance phases until the current phase is stable).

STATUS	MEANING
● UNLOCKED	Phase 1 (MVP) — available from Week 1–3 build
■ LOCKED — PHASE 2	Extended feature — unlocks after MVP (History/Repeater/Decoder) is stable
■ LOCKED — PHASE 3	Advanced / optional — unlocks after Phase 2 is stable

1. Top-Level Tab Bar (End-State)

The full tab bar across all phases. In Week 1 of the build, only the first four tabs render; the remaining tabs are reserved but not implemented and MUST NOT appear in the Phase 1 binary. This ordering reflects both build sequence and expected usage frequency.

FEATURE	BEHAVIOR	STATUS
[1] Proxy	Intercept on/off toggle, live traffic counter, port status.	UNLOCKED · P1
[2] History	Scrollable log of all requests/responses, detail split-pane, search/filter.	UNLOCKED · P1
[3] Repeater	Edit and resend any request; response comparison.	UNLOCKED · P1
[4] Decoder	Base64 / URL / HTML entity / hex encode-decode panel.	UNLOCKED · P1
[5] Intercept	Pause queue: forward/drop/edit requests before they leave the proxy.	LOCKED · P2
[6] Findings	Passive scanner results: missing headers, XSS sinks, open redirects.	LOCKED · P2
[7] Intruder	Payload-position fuzzing across a request template.	LOCKED · P3
[8] Scope	Host allow/deny rules controlling what gets intercepted.	LOCKED · P3
[9] Rules	Match & Replace — pattern-based auto request/response modification.	LOCKED · P3
[0] WebSocket	Frame-level inspection of WS traffic (currently passthrough).	LOCKED · P3

Navigation: **Tab** / **Shift+Tab** cycle tabs · number keys 1–0 jump directly to a tab (locked tabs show a toast: "Unlocks in Phase N — see roadmap") · **q** quits.

2. Phase 1 — MVP Tabs (Unlocked, Week 1–3)

These four tabs constitute the entire Week-1 TUI shell and remain the primary tabs for the life of the tool. All later phases add sub-panels to these tabs or new top-level tabs — they do not change this core structure.

2.1 Proxy Tab

Sub-panels / regions:

FEATURE	BEHAVIOR	STATUS
Status Bar	Shows listening port (default 8080), intercept ON/OFF, live request count.	UNLOCKED · P1
Toggle Control	Single keybind (i) flips intercept mode; visually highlighted when ON.	UNLOCKED · P1
Traffic Meter	Rolling counter: requests/sec, total bytes in/out for current session.	UNLOCKED · P1
CA Cert Notice	First-run banner: path to ca.crt and install instructions, dismissible.	UNLOCKED · P1
Pinned-Host Log	List of hosts that failed MITM due to certificate pinning (passthrough).	LOCKED · P2

2.2 History Tab

Sub-panels / regions:

FEATURE	BEHAVIOR	STATUS
Request List	Scrollable list: method, host, path, status, size, time. Color-coded by status class.	UNLOCKED · P1
Detail Split-Pane	Selecting an entry shows full request/response headers + body below/beside the list.	UNLOCKED · P1
Raw / Pretty Toggle	Switch body view between raw bytes and pretty-printed (JSON/XML/HTML).	UNLOCKED · P1
Search / Filter Bar	Filter by host, method, status code, or body substring.	LOCKED · P2
Persistent Store Badge	Small indicator showing SQLite is active vs in-memory-only mode.	LOCKED · P2
Findings Badge	Per-row icon if the passive scanner flagged an issue on that entry.	LOCKED · P2

Keymap — History Tab:

KEY	ACTION
↑ / ↓	Move selection in request list
Enter	Open detail pane for selected request
r	Send selected request to Repeater

/	Focus search/filter bar (Phase 2)
d	Send selected request/response body to Decoder

2.3 Repeater Tab

Sub-panels / regions:

FEATURE	BEHAVIOR	STATUS
Request Editor	Multiline editable raw request: method, path, headers, body.	UNLOCKED · P1
Send Control	Keybind/button opens raw TCP or TLS socket to original host and sends edited request.	UNLOCKED · P1
Response Viewer	Displays the new response: status, headers, body.	UNLOCKED · P1
Original vs New	Side-by-side (wide terminal) or stacked (narrow) comparison of original and new response.	UNLOCKED · P1
Response Diff Highlight	Highlights changed lines/bytes between original and replayed response.	LOCKED · P2
Save as Template	Store a Repeater request as a reusable named template.	LOCKED · P3

2.4 Decoder Tab

Sub-panels / regions:

FEATURE	BEHAVIOR	STATUS
Input Box	Free-text or pasted-in field for raw data to transform.	UNLOCKED · P1
Format Selector	Base64 / URL encoding / HTML entities / Hex.	UNLOCKED · P1
Encode / Decode Buttons	Explicit directional transform of the input.	UNLOCKED · P1
Chain Mode	Feed the output of one decode operation into the input of the next.	UNLOCKED · P1

3. Phase 2 — Extended Tabs (Locked until MVP is stable, Week 7–10)

Phase 2 introduces one genuinely new top-level tab (Findings) and one modal/queue overlay (Intercept), plus sub-panel additions to the four Phase 1 tabs already listed above.

3.1 Intercept Tab / Overlay

FEATURE	BEHAVIOR	STATUS
Pending Queue	List of requests currently paused awaiting user action.	LOCKED · P2
Edit-Before-Forward	Same editor UX as Repeater, applied to a live in-flight request.	LOCKED · P2
Forward Button	Sends the (possibly edited) request onward to the real target.	LOCKED · P2
Drop Button	Discards the request; nothing is sent to the target.	LOCKED · P2
Auto-Forward Rules	Optional: skip the queue for requests matching a scope/rule (depends on Phase 3 Scope tab).	LOCKED · P3

3.2 Findings Tab

FEATURE	BEHAVIOR	STATUS
Findings List	Flat list of scanner results: severity, host, rule triggered.	LOCKED · P2
Missing Security Headers	Flags absence of X-Frame-Options, X-Content-Type-Options, HSTS, CSP.	LOCKED · P2
Reflected Input Detection	Basic heuristic XSS sink flag: user input echoed unescaped in response body.	LOCKED · P2
Open Redirect Detection	Flags redirect responses whose Location header is attacker-influenceable.	LOCKED · P2
Jump-to-Request	Selecting a finding opens the originating entry in History.	LOCKED · P2

3.3 Storage & Export Additions

FEATURE	BEHAVIOR	STATUS
SQLite Persistence	All requests/responses written to data/history.db; survives restart.	LOCKED · P2
WAL Mode	Concurrent reads while proxy writes, avoids UI stalls on large sessions.	LOCKED · P2
Export as curl	Selected History entry can be copied/exported as an equivalent curl command.	LOCKED · P2

4. Phase 3 — Advanced Tabs (Locked until Phase 2 is stable, Week 10+)

Optional / later scope. These are the highest-complexity features and are the most likely to change shape once Phase 1–2 are in real use — treat this section as directional, not final.

4.1 Intruder Tab

FEATURE	BEHAVIOR	STATUS
Payload Position Marker	Mark one or more insertion points inside a base request template.	LOCKED · P3
Payload List Editor	Provide/paste a list of values to iterate through each marked position.	LOCKED · P3
Attack Runner	Executes the request once per payload, logging each response.	LOCKED · P3
Results Grid	Table of payload → status code → response length → response time, sortable.	LOCKED · P3

4.2 Scope Tab

FEATURE	BEHAVIOR	STATUS
Host Allow List	Only intercept traffic to/from listed hosts.	LOCKED · P3
Host Deny List	Explicitly exclude hosts even if broadly in scope (e.g. update servers).	LOCKED · P3
Wildcard / Regex Rules	Pattern-based scope matching, e.g. *.example.com.	LOCKED · P3

4.3 Rules Tab (Match & Replace)

FEATURE	BEHAVIOR	STATUS
Rule List	Ordered list of pattern → replacement rules applied automatically.	LOCKED · P3
Match Target	Choose to match against header, body, URL, or status line.	LOCKED · P3
Replace Action	Static replacement or simple templated value.	LOCKED · P3
Rule Enable/Disable Toggle	Per-rule on/off without deleting it.	LOCKED · P3

4.4 WebSocket Tab

FEATURE	BEHAVIOR	STATUS
---------	----------	--------

Frame List	Chronological list of WS frames per connection, direction indicated.	LOCKED · P3
Frame Detail	Raw payload viewer for a selected frame (text or binary/hex).	LOCKED · P3
Replay Frame	Resend a captured frame on an active connection (stretch goal).	LOCKED · P3

5. Global Navigation & Keymap

These bindings are consistent across every tab, whether unlocked or locked.

KEY	ACTION
Tab / Shift+Tab	Cycle forward / backward through unlocked top-level tabs
1-9, 0	Jump directly to a tab by its position in the bar
q	Quit BurpTUI
?	Open help overlay listing keys for the current tab
Esc	Close any open modal / cancel current edit

Locked tabs remain visible (dimmed) in the tab bar rather than hidden, so the end-state structure is always discoverable — pressing their number shows a one-line toast naming the phase that unlocks them. This keeps the roadmap visible inside the tool itself without cluttering the active UI with half-built features.

Design reference derived from the BurpTUI project spec (Phase 1–3 feature scope, Section: Feature Scope / Build Phases).